

Verilog and Hardware Descriptive Language

macOS Installation Guide

Since macOS is a Unix-based system, the cleanest and most reliable way to install these tools is via the terminal using Homebrew.

Step 1: Install Visual Studio Code & Extensions

1. Download and install **VS Code** from code.visualstudio.com.
2. Open VS Code, go to the Extensions tab (the square icon on the left), and search for **Verilog-HDL/SystemVerilog/Bluespec SystemVerilog** (by mshr-h). Install it for syntax highlighting.

Step 2: Install Homebrew (Package Manager)

If you don't already have Homebrew, open your **Terminal** and paste the following command, then press Enter:

```
Bash
```

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Step 3: Install Icarus Verilog (The Compiler/Simulator)

Once Homebrew is ready, run this command to install the **iverilog** and **vvp** tools:

```
Bash
```

```
brew install icarus-verilog
```

Step 4: Install GTKWave (The Waveform Viewer)

Because standard GTKWave is currently broken on macOS 14 (Sonoma) and later, use the patched community version built from the source code:

```
Bash
```

```
brew install --HEAD randomplum/gtkwave/gtkwave
```

Step 5: Verify the Mac Installation

Run these two commands in the terminal to ensure everything is recognized:

- `iverilog -V` (Should print the Icarus Verilog version)
- `gtkwave --version` (Should print the GTKWave version)

Windows Installation Guide

On Windows, the easiest method is to use a pre-compiled binary installer that conveniently bundles both Icarus Verilog and GTKWave together.

Step 1: Install Visual Studio Code & Extensions

1. Download and install **VS Code** from code.visualstudio.com.
2. Open VS Code, go to the Extensions tab, and install the **Verilog-HDL/SystemVerilog/Bluespec SystemVerilog** extension.

Step 2: Download the Icarus Verilog Installer

1. Go to the unofficial (but widely used as the standard) Windows binary page: bleyer.org/icarus
2. Download the latest `.exe` installer (usually the highest version number listed, e.g., `iverilog-v11-20201121-x64_setup.exe`).

Step 3: Run the Installer (Crucial Step)

1. Run the downloaded `.exe` file.
2. Accept the license agreement and proceed through the prompts.
3. **STOP** when you reach the "Select Additional Tasks" screen.
4. You **must check the boxes** that say:
 - **"Add executable folder(s) to the user PATH"**
 - **"Add executable folder(s) to the system PATH"**
(If you skip this, Windows terminal and VS Code will not know what `iverilog` or `gtkwave` are).
5. Click Next and Finish the installation. GTKWave is automatically installed alongside Icarus Verilog in this package.

Step 4: Verify the Windows Installation

1. Open the **Command Prompt** (`cmd`) or **PowerShell**. (If you had it open during the installation, close it and open a fresh one so the `PATH` updates).
2. Run `iverilog -V` to confirm the compiler is working.
3. Run `gtkwave` to confirm the waveform viewer opens successfully.